

Molecular Transport Phenomena in Amorphous Polyethylene Terephthalate and Polyethylene Furanoate Melts: Experimental Benchmarks and Computational Dynamics

Introduction to Gas and Vapor Transport in Polyesters

The mass transport of small penetrant molecules, specifically permanent gases like oxygen and highly interacting vapors like water, through polymer matrices represents a foundational area of study within polymer physics, materials science, and packaging engineering. For decades, polyethylene terephthalate has served as the ubiquitous industry standard for barrier applications, leveraging its favorable mechanical resilience, optical clarity, and relatively low gas permeability to dominate the global packaging landscape.¹ However, the escalating drive toward sustainable, bio-derived, and circularly economical materials has catalyzed intense academic and industrial research into polyethylene furanoate. Polyethylene furanoate is a furanic polyester analog synthesized from renewable biomass feedstocks, primarily utilizing 2,5-furandicarboxylic acid and ethylene glycol, positioning it as a direct, bio-based replacement for fossil-derived polyethylene terephthalate.¹

Extensive empirical evaluations across the literature have consistently demonstrated that polyethylene furanoate exhibits barrier properties that are markedly superior to those of polyethylene terephthalate. For instance, in the amorphous state at ambient conditions, polyethylene furanoate demonstrates an approximately eleven-fold reduction in oxygen permeability and a multifold reduction in water vapor diffusion compared to its terephthalate counterpart.⁵ Understanding the fundamental atomistic origins of these macroscopic transport phenomena requires a sophisticated, multiscale investigatory approach that tightly integrates macroscopic experimental benchmarking with high-fidelity molecular dynamics simulations.

For computational researchers and rheologists simulating polymer melts—such as highly parameterized systems comprising discrete simulation boxes of thirty-two chains of one hundred monomers—experimental data serve as the ultimate validation metric. Simulations conducted in the melt state, which typically occur at temperatures significantly above the glass transition temperature and the equilibrium melting temperature, require careful extrapolation from available experimental data. This is primarily because empirical data are predominantly gathered in the glassy amorphous or semi-crystalline states near ambient temperatures due to the severe physical instability, thermo-oxidative degradation, and rapid hydrolysis of polyesters at extreme elevated temperatures.⁸ This comprehensive report synthesizes the experimental

benchmarks for oxygen and water diffusivity in amorphous polyethylene terephthalate and polyethylene furanoate, characterizes the thermodynamic and kinetic activation energies governing these complex systems, and details the structural molecular mechanics responsible for the observed transport behaviors to provide a definitive framework for validating molecular dynamics melt simulations.

Theoretical Foundations of Mass Transport in Polymers

To effectively compare computational outputs derived from molecular dynamics trajectories with established experimental benchmarks, the fundamental thermodynamic and kinetic theories governing penetrant transport in dense polymer matrices must be rigorously established. The macroscopic permeability of a polymer film to a specific gas or vapor is mathematically defined by the classical solution-diffusion model. This model posits that permeability is the product of the thermodynamic solubility coefficient and the kinetic diffusion coefficient.

In this framework, the solubility coefficient describes the thermodynamic partitioning of the penetrant between the external phase and the polymer matrix, effectively quantifying the number of molecules that successfully dissolve into the polymer at equilibrium under a specific partial pressure. Conversely, the diffusion coefficient quantifies the kinetic mobility of these dissolved penetrant molecules as they traverse the internal free volume of the polymer matrix. Because the overarching differences in barrier performance between polyethylene furanoate and polyethylene terephthalate are driven almost entirely by profound disparities in their respective diffusion coefficients rather than their solubility coefficients, isolating and understanding the mechanics of diffusion is of paramount importance.⁶

Diffusion Kinetics and Fickian Transport Regimes

The transport of small molecules through a highly mobile polymer melt or an amorphous rubbery state is strictly governed by Fick's first law of diffusion. This fundamental principle postulates that the diffusive mass flux is directly proportional to the localized concentration gradient across the matrix. When analyzing experimental continuous-flow permeation data or pressure-decay sorption kinetics, researchers typically rely on Fick's second law to model the transient, time-dependent concentration profile before steady-state permeation is achieved.⁶

In the realm of molecular dynamics simulations, the diffusion coefficient is extracted directly from the atomistic trajectories of the penetrant molecules. This is achieved by calculating the mean squared displacement over time and applying the Einstein relation. The diffusion coefficient is determined from the linear, steady-state regime of the mean squared displacement. The transition from an initial period of anomalous, sub-diffusive behavior to the steady-state Fickian regime is highly dependent on the localized relaxation dynamics of the surrounding polymer chains. In the simulated melt state, the massively elevated thermal energy ensures that polymer chain segmental relaxations occur much faster than penetrant diffusion

events. This rapid structural turnover guarantees that the penetrant quickly enters the Fickian diffusion regime. Conversely, in the glassy state at lower temperatures, the rigid polymer backbone leads to extended periods of non-Fickian or anomalous diffusion.⁵ This is particularly true for highly interacting penetrants like water vapor, necessitating complex mathematical treatments such as the Berens-Hopfenberg structural relaxation model to isolate the purely diffusive component from the long-term structural swelling and matrix plasticization.⁷

Temperature Dependence and Arrhenius Activation Energy

Diffusion in dense polymer matrices is fundamentally a thermally activated rate process. The temperature dependence of the macroscopic diffusion coefficient is universally described by the Arrhenius equation. In this mathematical relationship, the diffusion coefficient at any given temperature is a function of a pre-exponential factor, the activation energy of diffusion, the universal gas constant, and the absolute temperature. The pre-exponential factor represents the theoretical maximum diffusion coefficient achievable at infinite temperature, essentially serving as a metric of the inherent entropic probability of a successful diffusive jump.¹⁵ The activation energy of diffusion quantifies the absolute energetic penalty associated with the localized separation of polymer chains required to accommodate the passage of the penetrant molecule.¹

For computational physicists conducting molecular dynamics simulations of polymer melts at elevated temperatures, such as five hundred to six hundred Kelvin, direct experimental validation relies entirely on a deep understanding of these Arrhenius parameters. Because physical experimental constraints prevent the direct measurement of gas diffusion in a pristine, un-degraded polymer melt, researchers must calculate the activation energy and the pre-exponential factor from empirical measurements taken at lower, stable temperatures. These empirically derived Arrhenius parameters are then utilized to mathematically extrapolate the expected diffusion coefficient up to the simulated melt conditions.¹⁶

The Free Volume Theory and Accessible Volume Convergence

The free volume theory of molecular diffusion provides the physical mechanism underlying the mathematical rate equations. This theory posits that penetrant transport does not occur via a continuous, fluid flow, but rather through discrete, instantaneous "hopping" events.⁵ A penetrant molecule resides within a localized microvoid, termed an accessible volume element, within the dense polymer matrix. It oscillates ballistically within this cage until a localized thermal fluctuation causes adjacent polymer chain segments to momentarily separate, creating a transient, sub-nanometer channel connecting the current void to a neighboring accessible volume element.⁵ The penetrant then executes a rapid translational jump through this transient channel.

Advanced computational methodologies, including grand canonical Monte Carlo insertion techniques coupled with long-timescale molecular dynamics trajectories, have revealed that the macroscopic displacement of oxygen and water relies entirely on the sporadic convergence of these accessible volume areas.⁵ The statistical frequency, spatial distribution,

and morphological evolution of these transient channels strictly dictate the magnitude of the macroscopic diffusion coefficient.¹⁸ Consequently, the relative thermomechanical rigidity and torsional resistance of the polymer backbone directly control the activation energy required to facilitate these critical volumetric convergences, establishing a direct link between polymer chain chemistry and macroscopic barrier performance.⁵

Methodological Bridges: From Molecular Dynamics to Macroscopic Reality

Translating the localized atomistic interactions observed in a molecular dynamics simulation into a macroscopic diffusion coefficient that can be accurately compared against experimental benchmarks requires rigorous computational protocols. For researchers simulating boxes consisting of thirty-two polymer chains, each containing one hundred repeating monomer units, the initial configuration and equilibration of the melt are of critical importance to ensure statistical validity and physical accuracy.²⁰

Simulation Box Configuration and Force Field Selection

A system size of thirty-two chains of one hundred monomers strikes an optimal balance in computational polymer rheology. This scale provides a sufficiently large bulk phase containing thirty-two hundred total monomeric units, which effectively mitigates artificial finite-size effects and self-interaction anomalies across periodic boundary conditions.²⁰ Simultaneously, this system size remains computationally tractable, allowing the simulation to proceed over the hundreds of nanoseconds required to capture the slow, rare-event diffusion dynamics of gases in dense aromatic polyesters.⁵

To accurately replicate the bulk densities and complex torsional energetics of amorphous polyethylene terephthalate and polyethylene furanoate, the selection of the interatomic potential, or force field, is paramount. The literature highlights the exceptional fidelity of highly parameterized, general-purpose all-atom force fields such as OPLS_2005 and AMBER in predicting the thermodynamic and kinetic behavior of these specific polyesters.²⁰ These force fields have been extensively validated against quantum mechanical density functional theory calculations, proving highly capable of replicating the precise energetic penalties associated with the rotation of phenyl and furan rings along the polymer backbone.⁵

System Equilibration and Penetrant Insertion

The process of generating a chemically accurate, completely amorphous polymer melt *in silico* requires a sequence of intensive thermodynamic equilibration steps. A standard, highly validated protocol begins by initializing the thirty-two polymer chains at an artificially low density to prevent catastrophic atomic overlap.²⁰ This expanded system is then subjected to isothermal-isobaric compression at elevated pressures and temperatures, often reaching one hundred atmospheres and seven hundred and fifty Kelvin, to rapidly compact the simulation

box to the approximate bulk density.²⁰

Following this initial compression, the system must be subjected to extensive simulated annealing cycles. This process involves repeatedly heating and cooling the melt between ambient temperature and highly elevated temperatures, such as twelve hundred Kelvin, over tens of nanoseconds. These thermal cycles provide the polymer chains with the necessary kinetic energy to overcome localized torsional barriers, allowing the one-hundred-monomer chains to relax into true, statistically uncorrelated Gaussian random coil conformations.²⁰ This vital step eliminates any artificial crystalline order, residual internal stress, or non-physical void spaces induced during the initial algorithmic packing phase, ensuring the resulting melt is a perfect representation of an isotropic amorphous fluid.²⁰

For the evaluation of diffusivity, penetrant molecules such as oxygen and water must be introduced into the equilibrated melt. Rather than randomly placing penetrants and risking severe localized atomic overlaps that could destabilize the simulation, researchers employ sophisticated stochastic insertion methods, such as Grand Canonical Monte Carlo algorithms.⁵ Furthermore, identifying optimal insertion sites is frequently achieved through the application of grid-scanning algorithms that map the internal potential energy surfaces of the simulation box. This scanning process locates preexisting pockets of accessible volume, allowing penetrants to be inserted in energetically favorable, chemically sensible locations.⁵ Once populated, a brief isothermal-isobaric relaxation phase allows the polymer matrix to swell infinitesimally to accommodate the partial molar volume of the inserted gases, transitioning the system into the final canonical production phase for data acquisition.⁵

Extracting Diffusivity from the Mean Squared Displacement

Extracting a statistically reliable diffusion coefficient from the molecular dynamics production phase requires an understanding of the temporal evolution of penetrant mobility. In dense polymer melts, penetrants exhibit three distinct transport phases over time. Initially, the molecules occupy a ballistic regime, characterized by rapid, unhindered vibration and translation within the confines of their immediate local microvoid.⁵ Because the molecules have not yet encountered the physical boundaries of the polymer chains, their mean squared displacement increases quadratically with time.

Following this brief initial phase, the penetrants enter an extended anomalous, or sub-diffusive, regime. In this phase, the molecules continuously collide with the surrounding polymer chains, remaining trapped within their localized cages while waiting for a rare thermal fluctuation to induce a chain rearrangement.⁵ The mean squared displacement curve flattens significantly during this period. Finally, at sufficiently long timescales, the system reaches the steady-state Fickian regime. In this phase, repeated volumetric convergences allow the penetrants to execute a series of sequential hopping events across multiple voids, yielding a strictly linear slope in the mean squared displacement over time.⁵

Because the diffusion coefficients in high-barrier polyesters like polyethylene terephthalate and polyethylene furanoate are inherently low, a single molecular dynamics trajectory often lacks

the statistical significance required to isolate the true Fickian regime from thermal noise. Consequently, robust computational methodologies require the execution of multiple independent replicate simulations—often generating five to fifteen unique configurations initialized with different velocity seed distributions.⁵ By monitoring penetrant diffusion over exceptionally long timescales, frequently up to two hundred nanoseconds per replica, researchers can calculate the ensemble average of the mean squared displacement. This averaging process smooths out localized kinetic anomalies and rare-event spikes, enabling the Einstein relation to accurately and reproducibly calculate diffusion coefficients in the range of ten to the negative eighth power square centimeters per second.⁵

Experimental Benchmarks: Oxygen Transport in Polyesters

The oxygen barrier performance of a polymer is universally regarded as a primary metric for its viability in commercial and industrial applications. The experimental quantification of oxygen diffusion in amorphous polyethylene terephthalate and polyethylene furanoate is typically executed utilizing highly sensitive, custom-built pressure-decay sorption systems or continuous-flow gas permeation cells.⁶ The data generated from these rigorous physical tests provide the definitive baseline against which molecular dynamics computational models must be calibrated and evaluated.

Empirical Oxygen Diffusion Coefficients

Extensive permeation testing conducted at standard benchmarking conditions, typically between two hundred and ninety-eight and three hundred and eight Kelvin under one atmosphere of oxygen partial pressure, has solidified the fundamental understanding of these materials. Empirical evidence universally confirms that polyethylene furanoate exhibits an order-of-magnitude reduction in oxygen permeability compared to standard polyethylene terephthalate.⁵ Critically, thermodynamic studies reveal that the overall solubility of oxygen in both amorphous polymer matrices is remarkably similar.⁶ Therefore, the dramatic eleven-fold reduction in macroscopic oxygen permeability observed in polyethylene furanoate is driven almost entirely by a corresponding and massive suppression of the kinetic diffusion coefficient.⁶

High-precision empirical measurements yield precise diffusion coefficients for these amorphous films, providing the necessary comparative benchmarks for computational studies.

Polymer Matrix System	Experimental Oxygen Diffusion Coefficient at 298 K (cm ² s ⁻¹)	Molecular Dynamics Predicted Oxygen Diffusion Coefficient (cm ²	Source References
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		s^{-1})	
Amorphous Polyethylene Terephthalate (PET)	$1.16 \times$	$3.24 \times$	⁵
Amorphous Polyethylene Furanoate (PEF)	1.04×10^{-9}	$2.88 \times$	⁵

The tight qualitative and quantitative agreement between the experimental benchmarks and the computational predictions generated from highly equilibrated molecular dynamics models strongly validates the utility of atomistic simulations in exploring complex transport phenomena.⁵ The minor overestimation that is typical of molecular dynamics predictions compared to experimental results is well understood within the literature. It is routinely attributed to the exceptionally high cooling rates applied during the *in silico* glass formation process, which mathematically locks in an excess amount of fractional free volume compared to the natural physical aging and slow relaxation experienced by real-world experimental samples.⁵ When researchers extrapolate these specific glassy-state values to the elevated temperatures of the melt state, they must apply mathematical adjustments utilizing established activation energies to ensure physical accuracy.

Thermodynamic Activation Energy of Oxygen Diffusion

The activation energy for oxygen diffusion serves to quantify the absolute energy penalty associated with the localized polymer chain rearrangements required to facilitate a successful penetrant jump.¹ The experimental derivation of this activation energy involves measuring the oxygen diffusion coefficient across a carefully controlled thermal window, and subsequently mapping the resulting data to extract the Arrhenius slope.¹

Empirical evaluations of oxygen transport reveal that the activation energy for oxygen diffusion in amorphous polyethylene furanoate is approximately 25 kilojoules per mole.²⁷ In parallel studies, the activation energy for oxygen diffusion in amorphous polyethylene terephthalate is estimated to be nearly identical, falling in the range of 24 to 25 kilojoules per mole for standard amorphous films.²⁷ While the overarching activation energy required for macroscopic oxygen transport remains highly similar between the two distinct polyesters, the specific energetic penalties governing individual chain-segment motions—which ultimately dictate the pre-exponential factor and the localized jump frequencies—diverge significantly, as will be discussed in subsequent sections.

For high-temperature molecular dynamics simulations of the polymer melt, the Arrhenius relationship serves as the crucial translational bridge. Because the melt state completely lacks

the rigid, frozen macroscopic structure of the glassy state, the entire diffusion mechanism shifts to a rubbery, highly liquid-like transport paradigm.¹⁷ The experimentally derived activation energies and their associated pre-exponential factors provide the fundamental mathematical constants required to confidently project the expected oxygen diffusion coefficient at simulated temperatures of five hundred or six hundred Kelvin. This mathematical projection is essential for ensuring that computational results derived from systems of thirty-two chains of one hundred monomers accurately reflect physical reality.¹⁶

Experimental Benchmarks: Water Vapor Transport in Polyesters

While oxygen serves as an ideal non-interacting permanent gas for evaluating pure physical free-volume effects, water vapor acts as a highly condensable, heavily interacting penetrant molecule. Water molecules possess the ability to form dynamic hydrogen bonds with the polar ester groups of polyethylene terephthalate and the highly polar furanic moieties of polyethylene furanoate. This interaction leads to exceptionally complex, concentration-dependent diffusion behavior, driving the plasticization of the polymer matrix, and resulting in complex dual-mode sorption kinetics.²⁹

Empirical Water Vapor Diffusion Coefficients

The diffusion of water vapor in amorphous polyethylene terephthalate and polyethylene furanoate has been meticulously categorized through advanced gravimetric sorption techniques. These methods utilize highly sensitive quartz crystal microbalances or dynamic vapor sorption instruments to record the transient, time-dependent uptake of water mass across varying humidity levels.¹⁴

At a standard testing temperature of thirty-five degrees Celsius, water diffusion within both amorphous polyethylene terephthalate and polyethylene furanoate maintains strictly Fickian characteristics up to a local water activity of approximately 0.6.⁷ Below this threshold, the transport is governed solely by the existing free volume of the dry polymer. However, beyond this critical activity threshold, the accumulation of water penetrants induces significant swelling and plasticization of the polymer architecture. This plasticization exponentially enhances localized polymer chain mobility, leading to a massive up-swing in the apparent diffusion coefficient alongside complex non-Fickian relaxation kinetics.⁷ To extract the fundamental, intrinsic diffusion coefficient under these highly interacting conditions, mathematical decoupling using the sophisticated Berens-Hopfenberg structural relaxation formulation is the standard empirical practice to separate classical Fickian diffusion from polymer chain relaxation.⁷

Experimental evaluations conducted at thirty-five degrees Celsius yield the following steady-state benchmark values for water vapor diffusivity in the strictly Fickian regime:

Polymer Matrix System	Experimental Water Vapor Diffusion Coefficient at 35 °C (cm ² s ⁻¹)	Average Reduction Factor Relative to PET	Source References
Amorphous Polyethylene Terephthalate (PET)	$\approx 1.5 \times$	Base Benchmark	³⁰
Amorphous Polyethylene Furanoate (PEF)	$\approx 2.6 \times$	$\approx$ Reduction	⁷

Averaged across the entire low-concentration Fickian interval, polyethylene furanoate exhibits a roughly five-fold reduction in the water diffusion coefficient compared to polyethylene terephthalate.⁷ However, it is vital for computational researchers simulating these systems to recognize that the overall thermodynamic solubility of water in polyethylene furanoate is significantly higher—approximately 1.7 times greater—than in polyethylene terephthalate.⁷ This increased solubility is driven by the enhanced polarity and hydrogen-bonding capacity of the furan ring compared to the non-polar phenyl ring.⁷ Therefore, the robust and superior overall water barrier property of polyethylene furanoate is driven exclusively by its highly suppressed kinetic diffusivity, which forcefully counteracts and overcomes the thermodynamic preference for water absorption.³³

Thermodynamic Activation Energy of Water Vapor Diffusion

Because the transport of water is intrinsically linked to the localized breaking and reforming of hydrogen bonds, as well as the concurrent physical displacement of the polymer chains, the activation energy parameters for water diffusion offer deep characterization of the polymer's viscoelastic state.³⁴ Experimental mapping of the water diffusion coefficients at intervals of fifteen, twenty-five, thirty-five, and forty-five degrees Celsius provides a highly precise Arrhenius interpretation of the transport mechanics.⁷

The empirical activation energies derived for water diffusion are remarkably identical between the two differing polyester matrices:

Polymer Matrix System	Activation Energy for Water Vapor Diffusion (kJ/mol)	Source References
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Amorphous Polyethylene Terephthalate (PET)	46.4 ±	7
Amorphous Polyethylene Furanoate (PEF)	47.1 ±	7

The statistical parity in these macroscopic activation energies underscores a fascinating physical reality: the absolute thermal energy required to initiate a successful diffusive water jump is nearly equivalent in both polyesters. The stark contrast in their actual, measured macroscopic diffusion rates must, therefore, be governed entirely by the pre-exponential factor and the statistical frequency of successful chain rearrangements, rather than a fundamental difference in the absolute energy barrier of the physical jump itself.⁷

The Free Volume Paradox and Atomistic Transport Mechanics

The persistent order-of-magnitude reductions in both oxygen and water diffusivity observed in polyethylene furanoate represent a significant and highly studied deviation from classical free volume theory, presenting a phenomenon often termed the "Free Volume Paradox" within the literature.¹¹ Experimental assessments utilizing advanced Positron Annihilation Lifetime Spectroscopy, as well as computational density evaluations, explicitly confirm that amorphous polyethylene furanoate actually possesses a higher fractional free volume than amorphous polyethylene terephthalate in the glassy state.³

Under classical diffusion paradigms, a higher fractional free volume should theoretically provide broader, more accommodating pathways for penetrant molecules, inevitably leading to elevated diffusion coefficients. The fact that polyethylene furanoate completely inverts this classical paradigm necessitates a rigorous, atomistic examination of the localized molecular mechanics governing the polymer backbone.³

Matrix Density and Physical Packing Architecture

To simulate precise and physically relevant diffusion metrics, the simulation box parameters defined in a molecular dynamics study must faithfully replicate the empirical densities of the target polymers. The amorphous solid-state densities strongly dictate the macroscopic volumetric limits of the system.

Polymer Matrix System	Experimental Bulk Density (g cm ⁻³)	MD Predicted Bulk Density at 298 K (g cm ⁻³)	Source References

Amorphous Polyethylene Terephthalate (PET)	1.559	1.503	5
Amorphous Polyethylene Furanoate (PEF)	1.562	1.547	5

While the absolute mass densities of polyethylene furanoate and polyethylene terephthalate are highly comparable, the internal architecture and physical distribution of that mass are profoundly different. The terephthalic acid monomer unit in polyethylene terephthalate features a perfectly linear, 180-degree para-substituted phenyl ring, which drives the polymer chains to adopt a predominantly extended zigzag conformation.³

Conversely, the 2,5-furandicarboxylic acid monomer in polyethylene furanoate exhibits an inherent, rigid non-linearity. The internal projection angle between the carbon bonds of the furan ring dictates a 129-degree geometric angle.³ Furthermore, the head-to-tail interatomic distance across the furan ring is 4.83 Angstroms, which is a full 16 percent shorter than the 5.73 Angstrom distance across the phenyl ring in polyethylene terephthalate.³ This severe geometric asymmetry forces polyethylene furanoate into a unique coiled-helix conformation in the amorphous state, alternating specifically between an anti-furan and gauche-ethylene glycol spatial arrangement.³ This distinct coiled architecture generates larger, isolated static microvoids—which directly accounts for the measured increase in overall fractional free volume—while simultaneously preventing the contiguous alignment of these voids into functional, long-range diffusion channels.³

Suppression of Dihedral Ring-Flipping Dynamics

The single most critical atomistic factor governing the kinetic suppression of diffusion in polyethylene furanoate is the severe restriction of segmental chain mobility, specifically the dampening of aromatic "ring-flipping" motions.⁵ In dense polymer melts and glasses, penetrant diffusion cannot occur through static, open pores; it fundamentally requires the active mechanical shifting of the polymer backbone to momentarily bridge adjacent free volume holes.⁵

In polyethylene terephthalate, the highly symmetric phenyl ring undergoes frequent, relatively low-energy 180-degree flips around its longitudinal axis. These highly active β -relaxation motions act as a dynamic, continuous gate, effectively sweeping and pushing penetrant molecules laterally through the matrix from one microvoid to the next.⁵

In stark contrast, this localized motion is massively hindered within the polyethylene furanoate matrix. The inherently asymmetric structure of the five-membered furan ring, combined with

the permanent dipole moment of 0.70 Debye emanating directly from the furanic oxygen heteroatom, imposes a severe energetic penalty on dihedral rotation.³ Extensive solid-state ^{13}C cross-polarization magic angle spinning nuclear magnetic resonance analyses, combined with high-level *ab initio* Density Functional Theory calculations, have definitively quantified these exact rotational energy barriers.⁵

Polymer Ring Moiety	Energy Barrier for Dihedral Ring-Flipping (kJ/mol)	Source References
Phenyl Ring (PET)	24.4	5
Furan Ring (PEF)	31.0	5

This roughly 27 percent increase in the fundamental energetic threshold required for dihedral rotation radically alters the temporal dynamics of the entire polymer matrix.⁵ The heightened rotational barrier essentially locks the furan rings in place, drastically reducing the statistical frequency at which the isolated static free volume domains can converge into functional transient channels.⁵

Consequently, the penetrant molecules—whether non-interacting oxygen or highly interacting water vapor—remain physically trapped within their localized microvoids for significantly extended durations.⁵ They experience only highly infrequent convergences that permit translational hopping.⁵ The dynamic, temporal rigidity of the coiled-helix structure completely negates the thermodynamic advantage of the increased fractional free volume, thereby establishing the definitive atomistic basis for the superior barrier properties of polyethylene furanoate.³

The Polymer Melt State: High-Temperature Extrapolations and Degradation Kinetics

For researchers computing diffusivity specifically within the polymer melt state—which represents the primary target of the thirty-two-chain, one-hundred-monomer simulation architecture under consideration—the reliance on classical experimental benchmarks introduces significant complexity. While molecular dynamics simulation environments offer mathematically perfect control over chemical stability at extreme temperatures, empirical measurements of gas diffusion at temperatures commensurate with the actual polymer melt are heavily compromised by the laws of chemical thermodynamics.

The Rarity of Experimental Melt Benchmarks

Polyethylene terephthalate exhibits a melting temperature of approximately 260 degrees

Celsius, while polyethylene furanoate melts near 211 degrees Celsius.³ Empirical efforts to capture physical, high-temperature diffusion profiles at or above these thresholds are routinely thwarted by rapid and catastrophic thermomechanical breakdown.⁸ At temperatures approaching 280 degrees Celsius, the introduction of even trace amounts of oxygen rapidly initiates severe thermo-oxidative degradation.⁹

Oxygen diffusion into the bulk polymer melt instantly facilitates the generation of highly reactive alkyl and peroxy macroradicals. These radicals aggressively attack the ethylene glycol linkages along the backbone.⁸ This aggressive oxidation cascade results in widespread random chain scission, driving a severe reduction in the overall molecular weight of the matrix and causing a rapid, corresponding drop in the measurable melt viscosity.⁸ Paradoxically, prolonged exposure to oxygen at these highly elevated temperatures can also trigger macromolecular crosslinking.⁸ The chemical coupling of adjacent macroradicals produces a three-dimensional gel network that artificially and significantly stiffens the melt. This gelation process fundamentally corrupts any concurrent attempts at diffusion measurement by drastically reducing the localized segmental mobility required for penetrant hopping.⁸

Similarly, the presence of trace water vapor at melt temperatures triggers catastrophic hydrolytic degradation. The immense kinetic energy of the high-temperature system, combined with the nucleophilic attack of water molecules on the polar ester bonds, rapidly cleaves the polymer chains.¹⁰ Because the activation energy threshold for hydrolysis is easily surpassed at 280 degrees Celsius, the structural integrity of a one-hundred-monomer chain is rapidly destroyed. Within minutes, the high-viscosity polymer melt is permanently transformed into a low-molecular-weight oligomeric fluid, rendering any physical diffusion measurement entirely unrepresentative of the intact polymer.⁴⁰

Consequently, the acquisition of purely experimental, uncorrupted oxygen and water diffusion coefficients within a pristine polyethylene terephthalate or polyethylene furanoate melt is physically restricted. The absolute highest-fidelity empirical data available for robust benchmarking are those captured rigorously at lower temperatures, typically 35 degrees Celsius, where the polymer remains perfectly stable in its glassy state.⁶ Therefore, advanced molecular dynamics simulation stands as the only pristine diagnostic tool capable of isolating pure penetrant diffusivity at 500 Kelvin to 600 Kelvin without the confounding, destructive variables of oxidative chain scission, crosslinking-induced gelation, or hydrolytic collapse.²⁵

Extrapolating the Arrhenius Relationship

To bridge the gap between empirical low-temperature data and the high-temperature computational environment, researchers must rely heavily on the mathematical principles of the Arrhenius relationship. When validating simulated melt diffusivities against experimental data gathered at 35 degrees Celsius, the theoretical shift in the transport mechanism across the glass transition temperature must be addressed.

In the rigid glassy state, the transport mechanism is best described by the dual-mode sorption

model, which mathematically accounts for both standard dissolution and the filling of static, frozen microvoids.²⁴ However, as the temperature exceeds the glass transition and enters the melt state, the infinite kinetic malleability of the polymer chains obliterates all static microvoids.²⁴ The capacity for localized hole-filling reduces to zero, and the system transitions entirely to an ideal Henry's law sorption paradigm.²⁴ Consequently, diffusion in the melt is purely Fickian, governed by the rapid, unhindered oscillation of the polymer backbone.

Experimental benchmarks obtained at 35 degrees Celsius must therefore be exponentially scaled using the Arrhenius activation energies (e.g., 25 kJ/mol for oxygen and 47 kJ/mol for water) to provide accurate mathematical targets for melt-state molecular dynamics outputs.¹ Utilizing these precise activation energies allows computational researchers to correctly project the expected diffusion coefficient up to 500 Kelvin, enabling a direct and valid comparison between the output of their thirty-two-chain simulation boxes and the physical reality dictated by empirical thermodynamics.¹

Synthesis and Strategic Outlook

The rigorous benchmarking of advanced molecular dynamics simulations governing oxygen and water vapor transport in polyethylene terephthalate and polyethylene furanoate demands strict alignment with a highly complex array of empirical observations and thermodynamic principles. A fully parameterized computational system, particularly one utilizing a large-scale architecture of thirty-two chains comprising one hundred monomers each, must successfully replicate the severe kinetic hindrance observed empirically in furanic matrices.²⁰

The experimental standards established across the literature clearly demonstrate that amorphous polyethylene furanoate delivers an eleven-fold reduction in oxygen diffusivity, measured at 1.04×10^{-9} square centimeters per second, and a five-fold reduction in water vapor diffusivity, measured at 2.6×10^{-9} square centimeters per second, compared to standard polyethylene terephthalate films at ambient conditions.⁵ These macroscopic barrier properties are inexorably linked to the localized atomistic molecular architecture of the specific polyester backbones. The substantial 31.0 kilojoule per mole energetic barrier that suppresses dihedral furan ring-flipping in polyethylene furanoate acts as a severe thermodynamic bottleneck.⁵ This restriction directly dampens the localized β -relaxations responsible for actively sweeping penetrant molecules through the matrix.⁵ Consequently, despite polyethylene furanoate harboring a measurably higher fractional free volume via its asymmetric coiled-helix conformation, the extreme infrequency of accessible volume convergences mathematically limits translational hopping.³

Furthermore, the empirically derived activation energy benchmarks for water vapor diffusion, which range narrowly from 46.4 to 47.1 kilojoules per mole, demonstrate near-perfect parity across both polymer matrices.⁷ This confirms that the ultimate distinction in transport efficiency relies primarily on the pre-exponential frequency factor dictated by dynamic chain

rigidity, rather than an absolute shift in the energetic requirement of the diffusive jump itself.⁷

Because pristine empirical measurements of gas diffusion at the true melt state, typically exceeding 260 degrees Celsius, are inevitably corrupted by rapid thermo-oxidative chain scission, massive crosslinking, and catastrophic hydrolysis⁸, computational researchers must carefully bridge the gap between low-temperature experimental benchmarks and high-temperature *in silico* environments. Relying on Arrhenius extrapolations generated from high-precision, non-degraded glassy-state data ensures that molecular dynamics simulations executed at 500 Kelvin or beyond remain firmly anchored to physical reality.¹ Integrating these multi-scale structural mechanics and kinetic realities provides a comprehensive, highly robust baseline for calibrating high-fidelity molecular models, directly advancing the computational predictive capacity for the next generation of sustainable, high-performance barrier polymers.

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